

In-hand cylinder reorientation with a delayed target

xArm7 + LEAP hand in MuJoCo 3.13. Scripted grasp, learned in-hand reorientation, target revealed at lift.

Result. The full task runs end to end under exact rules: contacts checked every 2 ms physics step, and URDF torque limits. It succeeds in **5.5 %** of 200 episodes (11/200, 95 % Wilson CI 3.1–9.6 %). The same grasp followed by holding still scores **4.0 %** (8/200, CI 2.0–7.7 %) on identical objects and targets. Eight of the policy's 11 successes happen *before* handover: the rod matches T^* during the wrist roll, exactly as in the baseline. The learned reorientation adds 3 (3/99 handovers; 10/64 on the validation seeds). The grasp loses half the episodes. This is far below the 50 % bar. What we can show is where and why it breaks.

A correction. An earlier version of this system reported 8.5 % end to end with 60 % handover (`results/v1_loose/`). That version sampled contacts every 40 ms and had no hand torque limit. Brief millimetre pops of the rod during the grasp fell between samples, so it was measuring a looser task than the spec. Everything below uses the exact rules.

Scope actually measured. Lying rods, diameter 4.4–5.6 cm, height 8–12 cm (α 1.4–2.7), density 500–900 kg/m³. Targets: axis within 40° of palm y, position in a 3 × 3 × 0.8 cm box on the palm. Outside that we report measured rates, not claims (§6).

System (same 200 episodes, seeds 123–126)	handover	success	success after handover	median e_{pos}	median e_{ang}
Scripted grasp + hold still (baseline)	49.5 %	4.0 %	0 / 99	2.9 cm	26°
Scripted grasp + learned reorient	49.5 %	5.5 %	3 / 99	2.9 cm	25°

Errors are at episode end over handed-over episodes. Success needs $e_{\text{pos}} \leq 1.5$ cm and $e_{\text{ang}} \leq 15^\circ$ held for 2 s, with the arm commanded fixed and hand-only contact (thresholds as specified, not loosened).

1. What we chose not to do, and why

- **One shape band, one start pose.** A one-day budget. Getting the full pipeline honest on one band beats a wide envelope we cannot measure. The wider envelope is swept in eval only (§6).
- **No learned grasp.** The grasp is a script. A learned grasp is the obvious next step. Its reward cannot see T^* (it is hidden until lift), so the right objective is the reorient critic's value averaged over the target prior. That is implemented (plan A in the repo) but not trained here.
- **No perception stack.** Ground-truth pose stands in for vision, gated by visibility (§2).
- **Plans A, B, C in the repo are not the submission.** They share this core but were not trained to convergence. Only the scripted-grasp + learned-reorient system (`Plan S`) is evaluated.

2. Assumptions

Item	Choice
Robot and limits	UFactory xArm7 + LEAP right hand (MuJoCo Menagerie), hand on a 40 mm solid bracket past the flange (a collision cylinder, r 3.2 cm, counted as an arm link). 23 position actuators with Menagerie gains; joint ranges from the models; all collisions on. Torque limits from the URDFs are enforced by the actuators: hand 0.95 N·m per joint (dexusuite <code>leap_hand_right.urdf</code> ; the Menagerie model has none), arm 50/30/20 N·m. Velocity limits (hand 8.48 rad/s, arm 3.14 rad/s) are not enforceable in a position servo, so every evaluated episode records its peak joint speeds and §7 reports them.
Palm frame	Origin on the palm face under the finger bases; x toward the fingertips, y across the fingers, z the outward palm normal.

Object	Uniform-density cylinder. r 2.2–2.8 cm, h 8–12 cm i.i.d., density 500–900 kg/m ³ (mass 60–270 g). Measured on the model with the hand open: index-to-ring fingertip span 9.1 cm, fingertip reach 13.6 cm from the palm origin. So the rod diameter is 0.48–0.62 of the span and its length 0.9–1.3 of the span.
Start pose	Lying on its side, uniform yaw, x 0.36–0.56 m, y $\pm$ 0.14 m in front of the robot base.
Friction	Object μ 0.7–1.1, ground μ 0.5–0.9, sampled per episode (sliding; MuJoCo default torsional/rolling).
Contact model / solver	2 ms step, implicit-fast, elliptic cones, impratio 10, 2 noslip iterations. Object geom condim 4, solref (10 ms, 1), solimp (0.9, 0.95, 2 mm). The 4 ms default was numerically stable but not accurate (§5). Contact resolution: object–ground contact counts when the surfaces are within 1 mm. This uses the floor's MuJoCo margin with an equal gap, so no force acts inside that band and the dynamics are unchanged. Without it, soft contacts that sink 1–3 mm flicker off for single 2 ms steps whenever a rod on the floor is disturbed, and each flicker would read as a lift followed by a violation. The band is stricter in both directions: lift requires the object at least 1 mm clear, and coming within 1 mm of the ground after lift is a violation.
Control	25 Hz, delta joint-position targets, no latency. The setpoint ramps linearly across the 20 physics steps of each control period, so commanded joint speed is at most 1.0 rad/s for the arm and 6.25 rad/s for the hand, both under the URDF limits. Grasp: arm and hand. Reorient: the arm holds the palm-up pose and fingers move at most 0.1 rad per step.
Tactile	Five pads (four fingertips, palm), read at 25 Hz: contact flag, summed normal force, force-weighted contact centroid in the pad frame. Noiseless. Resolution: one aggregate reading per pad (the sum over the pad's taxels, as a 4 × 4 Xela uSkin patch would give after summing). Counts every force-bearing contact, since a real pad does not know what it touches.
Cameras / vision gate	Three RealSense D435/D405-class RGB-D cameras, 640 × 480 at 30 Hz, noiseless and calibrated. Two are fixed at (1.1, $\pm$ 0.7, 0.9) m, aimed at (0.45, 0, 0.2) m, which covers both the floor workspace and the palm-up hold. One is on the wrist bracket, 11 cm behind and 6 cm above the palm face, aimed 2 cm above the palm origin. Field of view 87° × 58°, modelled conservatively as a cone of half-angle 29°. The pose-estimate stand-in updates at 5 Hz. On each update it exposes the ground-truth pose and (r , h) only if ≥ 30 % of 32 area-weighted surface samples lie inside some camera's field of view and are unoccluded from it (ray-cast; the cylinder occludes its own back side). Between updates, and when the gate fails, the last estimate is held and the flag reports whether the latest update passed. 30 % is about the silhouette a pose estimator needs for a textureless cylinder of known class; at handover the gate passes 93 % of the time.
Deployment observation	Arm/hand q and dq , previous action, current targets, tactile, palm pose from kinematics, gated vision, target block (zeros and flag 0 until lift), time. No object pose while occluded, no mass/friction, no object–ground forces.
Lift detector	After <i>every physics step</i> (2 ms), the object's contacts are classified by the other geom: hand, floor (with the 1 mm band), arm link (including the bracket), or other. Lift is the first physics step where the object touches ≥ 1 hand geom and nothing else. T^* enters the observation at the next control step. From the lift step on, any non-hand contact at any physics step ends the episode as a violation.
T^* sampling and feasibility	Drawn once per episode, fixed in the palm frame, independent of the grasp. Axis: palm y rotated by an angle uniform in $[0^\circ, 40^\circ]$ about a random perpendicular. Position: $x \in [-2.8, 0.2]$ cm, $y \in [-2.7, 0.3]$ cm, z = palm-contact height for that axis + $[0.2, 1.0]$ cm. Feasible = within 8 cm of the palm origin and no penetration with the palm/wrist at open fingers (resampled otherwise). The box was centred once on the median handover pose of the final grasp (300 handovers); the hold-still baseline measures how much of it is free.
Termination	Success; object contact with anything but the hand after lift (violation); 5 steps without hand contact after lift (drop); object > 0.5 m from the workspace centre; 20 s horizon.

Stage transition	Grasp → reorient when the lift flag is set <i>and</i> the arm has reached the palm-up hold configuration (joint error < 0.03 rad), or 200 steps after lift.
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3. Decomposition and stage interfaces

Stage 1: scripted grasp (`inhand/planc/scripted_grasp.py`). It reads only the deploy observation. The object pose comes from the gated vision stand-in, which sees the rod on the floor.

1. Move over the rod at a safe palm height of 25 cm. A straight joint-space move from home dips the fingers through the rod.
2. IK to a palm-down pre-grasp 10 cm above the rod: fingers across its axis, rod 5 cm toward the fingertips from the palm origin.
3. Descend until the palm is 2 cm *into* the rod top, which pins it to the floor.
4. Close the fingers over 100 steps (4 s).
5. **Catch:** if the lift flag flips during the close (the rod popped), finish the close and rise at 30 cm/s.
6. Lift 15 cm at 30 cm/s, then roll the wrist to palm-up in joint space.

It never reads T^* . It uses the lift flag only as an event.

Stage 2: learned reorient (`PlanS` in `inhand/systems.py`). A recurrent PPO actor on the deploy observation, running from handover to the end of the episode. The interface between the stages is the simulator state itself: the policy is trained from a bank of 765 states saved at real handovers (1,440 grasp episodes) (`scripts/collect_handovers.py`), mixed 70/30 with synthetic in-hand drops for coverage.

4. MDP of the learned component

Observation	Actor: the 161-d deploy vector above, with time measured from handover. Critic: deploy vector + object pose/velocity in the palm frame, r , h , mass, frictions, contact flags, true target (asymmetric actor-critic, train only).
Action	16 finger delta-targets in $[-1, 1] \times 0.1$ rad. Arm actions are zeroed: the target is palm-frame, and moving the arm only adds ways to touch the object with an arm link.
Reward	$\exp(-e_{\text{pos}}/3 \text{ cm}) + \exp(-e_{\text{ang}}/0.4 \text{ rad}) + 2 \cdot [\text{in tolerance}] + 50 \cdot [\text{success}] - 10 \cdot [\text{drop or violation}] - 0.002 \ a\ ^2$. All shaping is positive and bounded. With negative per-step error the first policies learned that ending the episode early (throwing the rod into the arm) was optimal.
Termination	As §2, 10 s horizon in training. In the full task the handover comes at about 11 s (median step 276 of 500), leaving about 9 s to reorient and hold.
Algorithm	PPO, GRU(256) over MLP(512, 256, 128). γ 0.99, λ 0.95, clip 0.2, 4 epochs $\times$ 4 minibatches, KL-adaptive LR (target 0.02), 56 envs $\times$ 64 steps per iteration.
Why	The fingers occlude the rod, so pose must be carried through time from tactile and proprioception: hence a recurrent actor. Privileged state makes the value function learnable in minutes rather than hours. Training directly on deploy observations removes the teacher–student distillation step, which the time budget could not afford.

5. The grasp stage: what it costs and what fixed it

The grasp decides both whether the episode reaches reorientation and where the object starts in the hand. Rates below are handovers (object hand-only after the palm-up roll) on lying rods of the trained band. The same scripted grasp went through two regimes. First it was tuned with contacts sampled every 40 ms and no hand torque limit. Then we checked every physics step with the URDF torque limits, and most of the apparent gains turned out to be sampling artefacts.

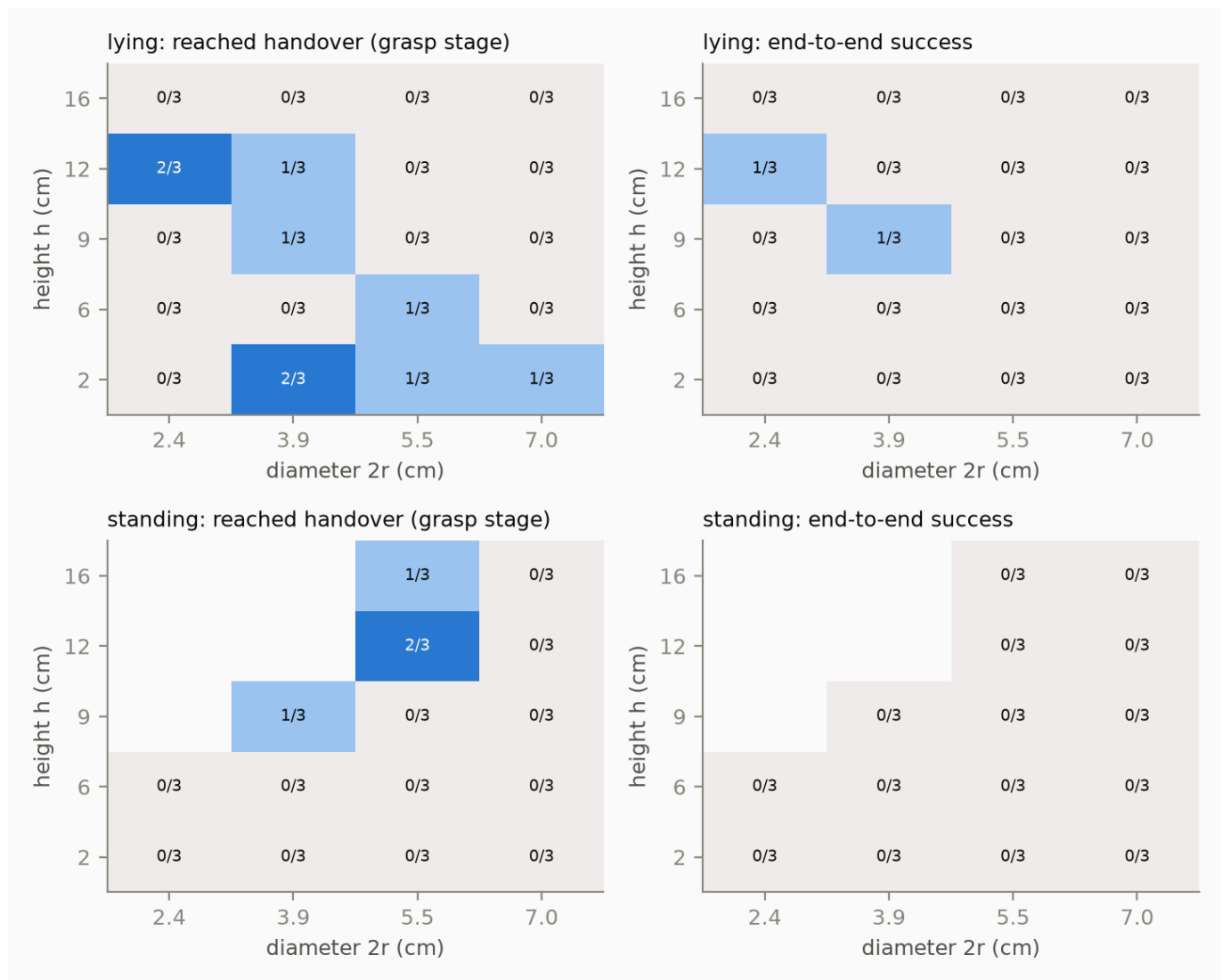
Setting	handover	Evidence
<i>Contacts sampled every 40 ms, no hand torque limit</i>		
Original script, 4 ms step, impratio 100, stiff object contact	1/12	Every failure: a finger flicks the rod at 0.5–0.8 m/s, which counts as lift, and it lands.
2 ms step, softer object contact, impratio 10	8/16	The flick was largely a time-step artefact (1 ms gave the same).
+ palm pressed into the rod, fast close, catch on lift flag	266/480	55 %. This was the first submission, with 8.5 % end to end.
<i>Every physics step, URDF torque limits (the submitted setting)</i>		
Same script	3–9 %	The close lifts the rod 1–2 mm for a few ms and it settles back. The 40 ms samples never saw it. Removing the torque limit did not help, so this is the detection, not the hand.
Slow close (4 s), no palm press, slower lift, standing rods	0–11 %	No press: the fingers lever the rod up during the close. Press: it springs out of the soft floor contact during the lift. Standing rods rock on an edge and flicker the same way.
+ 1 mm ground-contact resolution (§2)	22–25 %	Removes solver flicker. Also exposed a planning bug: the approach swept the fingers through the rod.
+ safe-height approach	21–26 of 64	No more knocks before the close.
+ fast lift (30 cm/s)	31/64	Slow lifts let the rod slip back onto the floor: a 1 mm/step start gave 10–15 of 64.
+ setpoints ramped across each 40 ms control period	47–59 %	Fixes a §3.3 problem: step setpoints drove the light fingers far past their 8.48 rad/s limit. The smoother close also pops the rod less.
Final script, 1,440 episodes	765 (53 %)	Saved as the reorient reset bank. Remaining failure: the rod slips back to the floor during the lift.

Trade-off. The scripted grasp needs no training, but it caps end-to-end success at about 53 %. Under exact contact rules its behaviour hinges on millimetre-scale effects that are hard to design by hand. It also hands over a spread of in-hand poses (axis 1°–47° from palm y, 10th–90th percentile) that the reorient policy must absorb. A learned grasp trained under the same exact rules is the change with the largest expected gain. Its reward cannot see T^* before lift, so the right objective is the reorient critic's value averaged over the target prior (plan A in the repo, not trained here).

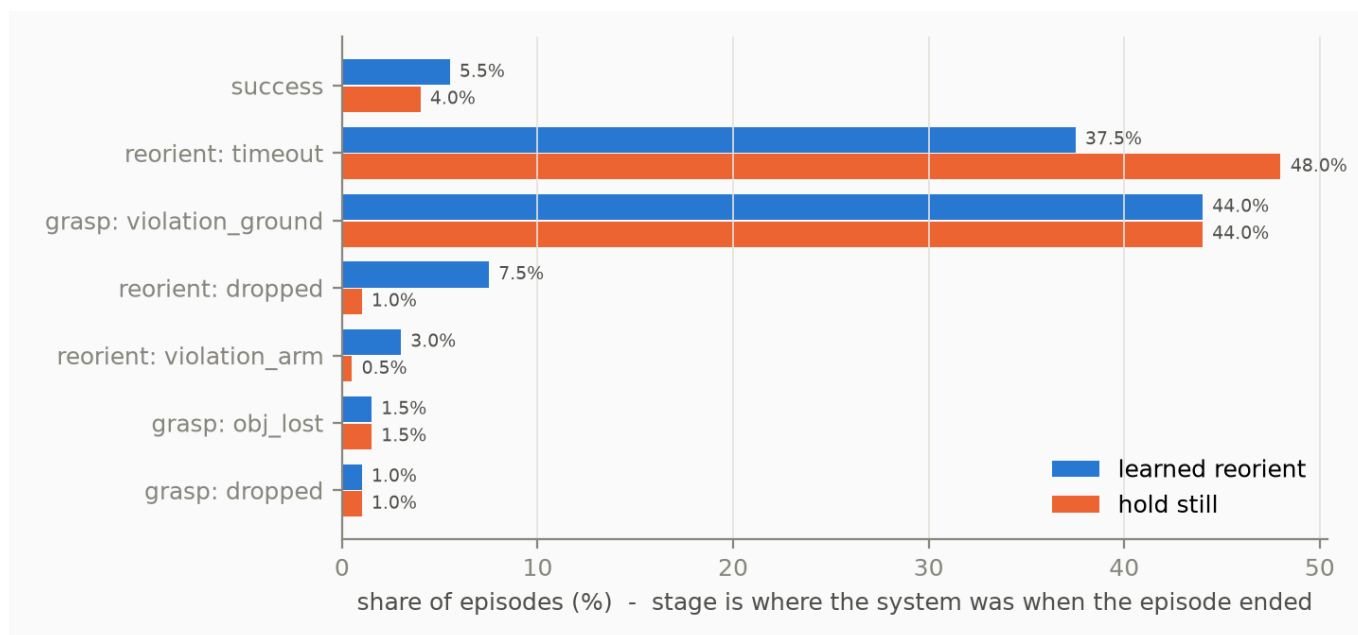
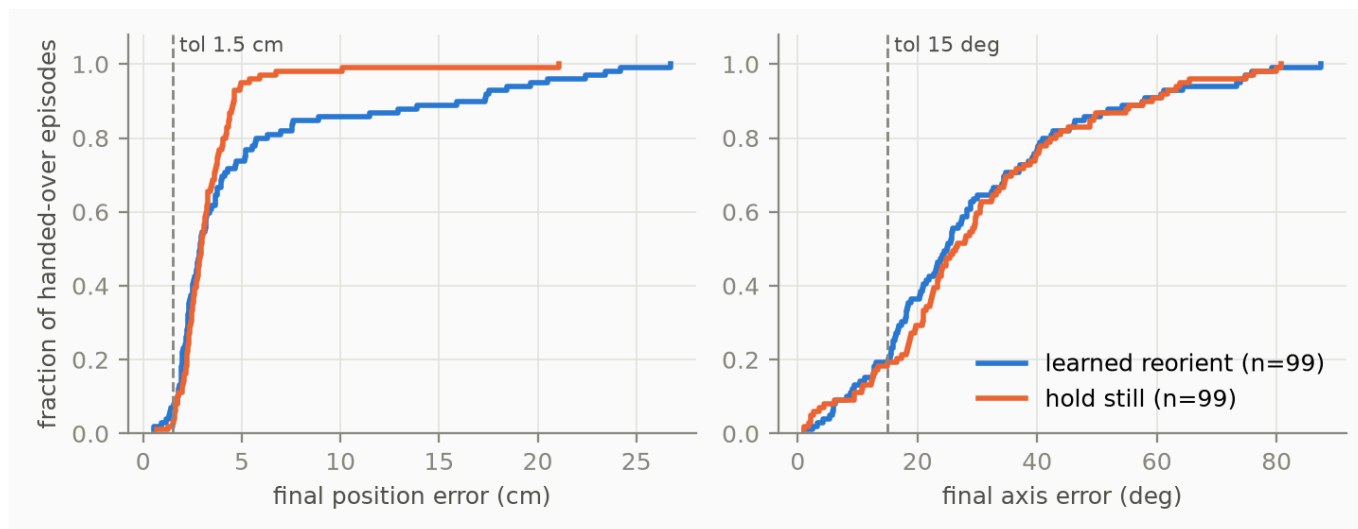
6. Envelope

The policy was trained on the narrow band only. We swept the WIDE envelope in eval: diameter 2.4–7.0 cm and height 2–16 cm on a 4×5 grid, 3 objects per cell, both start poses (standing only where $\alpha \leq 3$; tall standing rods are tipped over first). 105 episodes.

- **Grasp.** Handover in 12 % overall: lying 9/60, standing 4/45, never more than 2/3 in any cell. The rod popping and settling back to the floor is the dominant failure at every diameter (16/21 at 2.4 cm, 14/24 at 3.9 cm, 23/30 at 5.5 cm, 23/30 at 7.0 cm). Fat rods are also pushed out of the workspace (6/30).
- **End-to-end.** 2/105 (1.9 %): a lying 2.4×12 cm rod after handover, and a lying 3.9×9 cm rod during the roll. Three objects per cell cannot tell these apart from luck.
- **Measured range.** Lying rods, diameter 4.4–5.6 cm, height 8–12 cm (α 1.4–2.7), 5.5 % success, of which reorientation adds 1.5 points. We claim no other diameter or start pose.



7. Errors and failure taxonomy



Rates over the 200 paired episodes, attributed to the stage the system was in when the episode ended.

Failure	Stage	learned	hold	Why
Object touches the floor after lift	grasp	44.0 %	44.0 %	The close or the lift pops the rod a millimetre or more off the floor (this is lift) and it settles back. Identical in both columns because the grasp is shared.
Timeout, in hand but out of tolerance	reorient	37.5 %	48.0 %	The policy rarely moves the rod far enough. Of handed-over episodes, position ends within tolerance in 7 % (hold 2 %), axis in 19 % (hold 18 %), both in 4 % (hold 0 %).
Dropped (5 steps without hand contact)	reorient	7.5 %	1.0 %	Caused by the policy: moving the rod with torque-limited fingers costs grip.
Object touches an arm link or the bracket	reorient	3.0 %	0.5 %	Caused by the policy: rolling the rod toward the wrist.
Lost or dropped during the grasp	grasp	2.5 %	2.5 %	Pushed away before the close, or slipped during the roll.

Success	5.5 %	4.0 %	8 of the 11 learned successes, and all 8 baseline successes, happen during the roll: the arm freezes when the rod passes through tolerance and it holds for 2 s.
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Per-stage attribution: the grasp stage loses 46.5 % of episodes, the reorient stage 48.0 %, and 5.5 % succeed. Removing the grasp pop would roughly double the number of episodes the policy sees; the policy itself is the weaker link once there.

Joint limits. Joint ranges and torques are within limits by construction. Commanded speeds stay under the URDF limits (§2). Measured arm speed never exceeded 3.14 rad/s in these 200 episodes (peak 2.5). Measured hand speed exceeded 8.48 rad/s in 3.6 % of control steps (93 % of episodes, peak 65 rad/s). In 44 % of those steps the actuator was at its 0.95 N·m limit. In a 6-episode diagnostic, every over-limit sample happened with the hand in contact with the object or the floor, during the lift or roll, and none in free motion. These are back-driven contact transients, not commanded motion. The Menagerie LEAP model has no gearbox inertia to resist them, and actuator dynamics are out of scope (§4.1). We report them rather than hide them.

8. Limitations and next steps

- **The grasp is the ceiling, and it is fragile under exact rules.** Half of all episodes end before reorientation. A scripted top-down grasp of a lying rod keeps lifting it by a millimetre or two for a few ms. Next: train the value-aware learned grasp (plan A) directly under the exact rules, warm-started from the script. Its reward cannot see T^* , so it maximises the reorient critic's value averaged over the target prior.
- **The reorient policy barely beats holding still.** 3/99 after handover on the test seeds (10/64 on validation), plus more drops than the baseline. It trained for 80 min (8.6 M steps) on a CPU. Next: more steps (MJX or MuJoCo Warp on the GPU cluster), a target-cone curriculum from 10° outward, and tactile-driven regrasp primitives.
- **Target prior centred on the grasp's typical pose.** A declared feasibility choice. The hold-still baseline shows how much it gives away for free (4 %, all during the roll).
- **Contact resolution.** The 1 mm ground band is a modelling decision, stricter than exact touch in both directions, but a reviewer may prefer a different band. `scene.CONTACT_RESOLUTION` is one constant.
- **Passive over-speed transients** in the hand (§7). A gearbox inertia (armature) matched to the XC330 servos would be the next modelling step if actuator dynamics came into scope.
- **Narrow band.** One shape band, one start pose. The wide sweep in §6 shows nothing outside it works reliably.
- **Vision stand-in.** Ground truth when ≥ 30 % visible. A real estimator (e.g. FoundationPose) would add latency and error we have not modelled.

9. Compute

All on one laptop CPU (Apple M5 Max, 18 cores), CPU MuJoCo, no GPU.

- **Submitted reorient policy:** 80 min, 56 envs over 14 processes, 8.61 M env steps at 1,790 steps/s. The submitted checkpoint is the best-smoothed-success snapshot at 5.59 M steps (52 min). It was chosen over the final checkpoint on 120 held-out validation episodes (seeds 900–903): 12/120 vs 8/120. The test seeds (123–126) were not used for selection.
- **Superseded runs:** about 8.8 M steps in total, including the first 80-min run under the loose 40 ms rules.
- **Grasp sweeps:** about 1 h in total on up to 16 processes, 12–32 episodes per setting.
- **Handover banks:** about 3 min on 16 processes. **Evaluation:** about 5 min on 12 processes.
- **Cluster:** NCSA Delta jobs were queued as a hedge, never started, and were cancelled.

Total env steps for training: about 17.4 M. Active wall clock: about 4.5 h over one day.

10. References and tools

MuJoCo 3.13 and MuJoCo Menagerie (xArm7, LEAP hand); PyTorch; custom PPO following rl_games' KL-adaptive schedule. Pinto et al., Asymmetric Actor Critic (2017); Chen et al., Visual Dexterity (2023); Handa et al., DeXtreme (2023); Andrychowicz et al., HER (2017); Ross et al., DAgger (2011).

Appendix: requirement checklist

Spec	How it is met	Where
§2.1 fixed-base arm + multi-finger hand, infinite ground at $z = 0$, gravity	xArm7 + LEAP (4 fingers) on a MuJoCo plane; gravity default -9.81 .	<code>inhand/scene.py</code>
§2.1 randomized stable pose, position, yaw, friction	Lying (and standing where $\alpha \leq 3$, eval only), uniform position and yaw in the workspace, settled 0.2 s; friction sampled per episode.	<code>inhand/cylinder.py</code> , <code>config.Friction</code>
§2.2 (r, h) i.i.d. from declared G; density; scale vs hand	Uniform, independent; §2 table gives bounds against the measured span and reach.	<code>config.Envelope</code> , <code>config.WIDE</code>
§2.3 palm frame; e_{pos} , $e_{\text{ang}} = \arccos \hat{a} \cdot \hat{a}^* $	Defined in §2; errors ignore roll and end flip.	<code>inhand/frames.py</code> <code>errors</code>
§2.4 1.5 cm, 15° , hand-only, 2 s hold with the arm commanded fixed	Thresholds unchanged. Arm setpoints are frozen from the first in-tolerance step. The hold counter resets on leaving tolerance; any violation or drop ends the episode.	<code>env._signals</code> , <code>config.Success</code>
§2.4 full error distribution	Every episode's final errors are saved; CDFs and percentiles in §7.	<code>results/*.json</code>
§2.5 T^* hidden until lift, including from the grasp and the stage selector	The deploy target block is zeros with flag 0 until lift. The grasp and stage switch read only the flag (an event), never the target values.	<code>env._obs</code> , <code>systems.PlanS</code> , <code>test</code> <code>test_target_hidden_until_lift</code>
§2.5 lift = first time hand-only; after lift hand-only or fail	Checked after every 2 ms physics step, with a 1 mm ground-contact band (stricter both ways, §2).	<code>env.step</code> , <code>test</code> <code>test_brief_hand_only_contact_counts_as_lift</code>
§3.1 only hardware-obtainable signals	Deploy vector: proprioception, action history, tactile, kinematics, gated 5 Hz vision stand-in, target after lift. Privileged state goes to the critic only. Deployed actors are rejected unless trained on the deploy vector.	<code>inhand/observations.py</code> , <code>rl/ppo.load_actor</code>
§3.1 tactile layout, resolution, rate	5 pads, aggregate per pad, 25 Hz (§2).	<code>contacts.tactile</code>
§3.1 cameras: rate, pose, resolution; vision gate	3 cameras, poses and aims, 640×480 , $87^\circ \times 58^\circ$ field of view, 5 Hz estimate. The gate is $\geq 30\%$ of the surface in view and unoccluded.	<code>config.Sensors</code> , <code>contacts.visible_fraction</code> , <code>test</code> <code>test_vision_updates_only_on_camera_frames</code>
§3.2 explicit train/deploy split	DEPLOY vs PRIV layouts; <code>actor_key</code> in checkpoints; input-width assertion.	<code>observations.py</code> , <code>systems.ActorRunner</code>
§3.3 torque, velocity, joint limits; collisions on	Joint ranges from the models. Torque limits enforced (hand 0.95 N·m from the URDF). Commanded speeds under the URDF limits. Measured peaks	<code>scene.py</code> , <code>eval.py</code> <code>speed_limits</code> , <code>test</code> <code>test_hand_torque_is_limited_to_the_urdf</code>

reported (§7). No collisions disabled;
the bracket was added as a solid.

§4.2 decomposition and switching rule	§3.	<code>systems.PlanS</code>
§4.3 assumptions	§2 table.	<code>inhand/config.py</code>
§5 envelope: α , r and h vs hand, pose class	§6: trained band plus a measured wide sweep.	<code>results/s_grid_wide.json</code>
§6.1 repo, README with exact commands	README "Reproduce" lists the commands run, in order.	<code>README.md</code>
§6.2 checkpoint + single eval.py	<code>python eval.py</code> loads <code>checkpoints/reorient_deploy.pt</code> and runs the submitted system.	<code>eval.py</code>
§6.3 report sections	§1–§9.	this document
§6.4 videos	Successes and failures across the trained band and the wide envelope.	<code>videos/</code>
§8 compute stated	§9.	
§1 end-to-end success above 50 %	Not met: see the headline and §7.	